2o2dy

File

Name: P3-K02-helical
Description: KISSsoft example
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	12.500	
Speed (1/min)	[n]	2950.0	765.9
Number of load cycles (in mio.)	[NL]	3540.000	919.038
Torque (Nm)	[T]	40.5	155.9
Application factor	[KA]	1.25	
Required service life (h)	[H]	20000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	101.845	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	1.5000	
Transverse module (mm)	[mt]	1.5459	
Normal diametral pitch (1/in)	[Pnd]	16.93333	
Transverse diametral pitch (1/in)	[Ptd]	16.43034	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	14.0000	
Number of teeth	[z]	27	104
Facewidth (mm)	[b]	21.00	20.00
Hand of gear		left	right
Accuracy grade	[Q-DIN 3961:1978]	6	6
Inner diameter (mm)	[di]	0.00	146.39
Shaft diameter (mm)	[ds]	0.00	146.39
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1
31CrMoV9, Nitriding steel, gas-nitrided, ISO 6336-5 Figure 13a/14a (MQ)

Gear 2
42CrMo4 (3), Through hardened steel, nitrided, ISO 6336-5 Figure 13b/14b (MQ)

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HV 800	HV 550
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	420.00	363.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1250.00	998.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1100.00	1100.00
Yield point (N/mm²)	[σS]	900.00	900.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	3.00	3.00
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	20.00	20.00
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	Mobilgear 600 XP 150	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	150.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	14.70
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	75.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	21.358
(mm)	[a0.e/i]	21.298 /21.270
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.606 / 0.608

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000

Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	80.499
(mm)	[a0.e/i]	80.389 /80.320
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.591 / 0.593

4.1.1 Information on final machining

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	for high load capacity gearboxes		
Tip relief (µm)	[Ca L/R]	5.0 / 5.0	5.0 / 5.0

4.2 Basic data

Overall transmission ratio	[itot]	-3.852
Gear ratio	[u]	3.852
Transverse module (mm)	[mt]	1.546
Transverse pressure angle (°)	[αt]	20.562
Working pressure angle (°)	[αwt]	21.425
(°)	[αwt.e/i]	21.450 /21.400
Working pressure angle at normal section (°)	[αwn]	20.838
Base helix angle (°)	[βb]	13.140
Helix angle at operating pitch circle (°)	[βw]	14.078
Reference center distance (mm)	[ad]	101.258
Sum of profile shift coefficients	[Σxi]	0.3994

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient	[x]	0.3255	0.0739
Generating profile shift coefficient	[xE.e/i]	0.2852/ 0.2669	0.0006/-0.0452
Virtual gear no. of teeth	[zn]	29.343	113.025
Involute length (mm)	[l_dFa-l_dFf]	3.079	3.104

4.3 Diameters and their allowances

		------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)	[d]	41.740	160.776
Base diameter (mm)	[db]	39.081	150.533
Tip alteration (mm)	[k*mn]	-0.012	-0.012
Tip diameter (mm)	[da]	45.692	163.973
(mm)	[da.e/i]	45.692 /45.682	163.973 /163.963
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.010	0.000 / -0.010
Tip form diameter (mm)	[dFa]	45.692	163.973
(mm)	[dFa.e/i]	45.692 /45.682	163.973 /163.963
Active tip diameter (mm)	[dNa]	45.692	163.973
(mm)	[dNa.e/i]	45.692 /45.682	163.973 /163.963
V-Circle diameter (mm)	[dv]	42.716	160.997
(mm)	[dv.e/i]	42.595 /42.541	160.778 /160.640

Operating pitch diameter (mm)	[dw]	41.982	161.708
(mm)	[dw.e/i]	41.989 /41.975	161.736 /161.680
Root diameter (mm)	[df]	38.966	157.247
(mm)	[df.e/i]	38.845 /38.791	157.028 /156.890
Active root diameter (mm)	[dNf]	40.193	158.852
(mm)	[dNf.e/i]	40.221 /40.171	158.888 /158.821
Root form diameter (mm)	[dFf]	40.081	158.171
(mm)	[dFf.e/i]	40.006 /39.973	157.980 /157.862

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.375	0.375
Effective tip clearance (mm)	[c.e/i]	0.576 / 0.468	0.486 / 0.418
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.124 / 0.082	0.513 / 0.421
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	1.976	1.599
(mm)	[ha.e/i]	1.976 / 1.971	1.599 / 1.594
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	1.387	1.764
(mm)	[hf.e/i]	1.447 / 1.475	1.874 / 1.943
Tooth height (mm)	[h]	3.363	3.363

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	34.709 /34.681	24.746 /24.736
Roll angle to dNf (°)	[ξNf.e/i]	13.943 /13.626	19.353 /19.272
Roll angle at dFf (°)	[ξFf.e/i]	12.541 /12.312	18.243 /18.094

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8077	1.6246
Normal tooth thickness at tip circle (mm)	[san]	0.973	1.218
(mm)	[san.e/i]	0.931 / 0.903	1.141 / 1.086
Normal tooth thickness at tip form circle (mm)	[sFan]	0.973	1.218
(mm)	[sFan.e/i]	0.931 / 0.903	1.141 / 1.086
Normal space width at root circle (mm)	[efn]	0.000	1.077
(mm)	[efn.e/i]	0.000 / 0.000	1.090 / 1.099
Pitch on reference circle (mm)	[pt]	4.857	4.857
Base pitch (mm)	[pbt]	4.547	4.547
Transverse pitch on contact-path (mm)	[pet]	4.547	4.547
Lead height (mm)	[pz]	525.932	2025.813
Axial pitch (mm)	[px]	19.479	19.479

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	1.622	1.157
Specific sliding at the tip	[ζa]	0.444	0.444

Specific sliding at the root	[ζf]	-0.798	-0.798
Mean specific sliding	[ζm]	0.444	
Sliding factor on tip	[Kga]	0.250	0.178
Sliding factor on root	[Kgf]	-0.178	-0.250

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	31.954
Transverse contact ratio	[εα]	1.571
	[εα.e/m/i]	1.581 / 1.568/ 1.555
Overlap ratio	[εβ]	1.027
Total contact ratio	[εγ]	2.597
	[εγ.e/m/i]	2.608 / 2.595/ 2.582
Length of path of contact (mm)	[gα]	7.142
(mm)	[gα.e/m/i]	7.190 / 7.131/ 7.072
Approach path of contact (mm)	[gf]	2.973
(mm)	[gf.e/m/i]	3.011 / 2.967/ 2.922
Recess path of contact (mm)	[ga]	4.170
(mm)	[ga.e/m/i]	4.179 / 4.165/ 4.150

		Gear 1	Gear 2
Addendum contact ratio	[ε]	0.917	0.654
	[ε.e/i]	0.919 / 0.913	0.662 / 0.643
Length T1-A and T2-A (mm)	[T1A,T2A]	4.695	32.508
(mm)	[.e/i]	4.647 / 4.755	32.508 / 32.495
Length T1-B and T2-B (mm)	[T1B,T2B]	7.290	29.912
(mm)	[.e/i]	7.290 / 7.280	29.864 / 29.970
Length T1-C and T2-C (mm)	[T1C,T2C]	7.668	29.535
(mm)	[.e/i]	7.658 / 7.678	29.497 / 29.573
Length T1-D and T2-D (mm)	[T1D,T2D]	9.242	27.960
(mm)	[.e/i]	9.194 / 9.303	27.960 / 27.948
Length T1-E and T2-E (mm)	[T1E,T2E]	11.837	25.365
(mm)	[.e/i]	11.837 / 11.828	25.317 / 25.423
Length T1-T2 (mm)	[T1T2]	37.202	
(mm)	[.e/i]	37.154 / 37.250	
Diameter of single contact point B (mm)	[d-B]	41.712	161.986
(mm)	[d-B.e/i]	41.712 / 41.705	161.950 /162.028
Diameter of single contact point D (mm)	[d-D]	43.232	160.585
(mm)	[d-D.e/i]	43.191 / 43.283	160.585 /160.576

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	1938.82	
Axial force (N)	[Fa]	483.40	483.40
Radial force (N)	[Fr]	727.28	727.28
Normal force (N)	[Fnorm]	2126.42	2126.42
Nominal circumferential force per mm (N/mm)	[w]	96.94	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	1927.65	

Axial force (N)	[Faw]	483.40		483.40
Radial force (N)	[Frw]		756.41	
Circumferential speed reference circle (m/s)	[v]		6.45	
Circumferential speed operating pitch circle (m/s)	[v(dw)]		6.48	

5.2 Contact stiffness

Running-in value (µm)	[yp]		0.6	
Running-in value (µm)	[yf]		0.4	
Tolerances f_{pe}, f_f and $f_{H\beta}$, matching the tolerances in section 7				
Correction factor	[CM]		0.800	
Gear blank factor	[CR, bs/b, sr/mn]		0.931	(0.500,3.500)
Basic rack factor	[CB]		0.975	
Material coefficient	[E/Est]		1.000	
Single stiffness (N/mm/µm)	[c']		13.625	
Theoretical single stiffness (N/mm/µm)	[c'th]		19.334	
Contact stiffness (N/mm/µm)	[cγα]		19.458	
Contact stiffness (N/mm/µm)	[cγβ]		16.539	
Reduced mass (kg/mm)	[mRed]		0.00525	
Resonance speed (min-1)	[nE1]		21531	
Resonance ratio (-)	[N]		0.137	
Subcritical range				

5.3 Calculation of K factors

Running-in value (µm)	[yα]		0.3	
Bearing distance l of pinion shaft (mm)	[l]		42.000	
Distance s of pinion shaft (mm)	[s]		4.200	
Outside diameter of pinion shaft (mm)	[dsh]		21.000	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]			4
0:a), 1:b), 2:c), 3:d), 4:e)				
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']			-1.00
Without stiffening				
Effective flank line deviation (µm)	[Fβy]		4.64	
from deformation of shaft (µm)	[fsh·B1]		0.40	
fsh (µm)	[fsh]		0.81	
B1	[B1]		0.50	
fHβ5 (µm)	[fHβ5]		6.00	
Flank line:	crowned [$C_\beta = 0.5 \cdot (f_{m\alpha}+f_{sh})$]			
Position of contact pattern:	favorable			
from production tolerances (µm)	[fmα*B2]		6.02	
B_2=0.50				
Tooth trace deviation, theoretical (µm)	[Fβx]		5.46	
Running-in value (µm)	[yβ]		0.82	

5.4 K factors

Dynamic factor	[Kv]		1.070
Face load factors			
- Flank	[KHβ]		1.296
- Tooth root	[KFβ]		1.242
- Scuffing	[KBβ]		1.296
Transverse load factors			
- Flank	[KHα]		1.090
- Tooth root	[KFα]		1.090
- Scuffing	[KBα]		1.090

Application factor	[KA]	1.250
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.2669	-0.0452
Tooth form factor	[YF]	0.953	1.101
Stress correction factor	[YS]	2.164	2.073
Load application angle (°)	[αFen]	21.301	20.285
Load distribution influence factor	[fε]	0.777	
Load application diameter, virtual spur gear (mm)	[d_{en}]	45.301	170.573
Load application diameter (mm)	[d_{en}]	43.027	161.811
Bending moment arm (mm)	[hFe]	1.456	1.815
(-)	[hFe*]	0.970	1.210
Tooth root thickness at critical cross section (mm)	[sFn]	3.255	3.392
(-)	[sFn*]	2.170	2.261
Root fillet radius at critical cross section (mm)	[ρF]	0.707	0.701
(-)	[ρF*]	0.471	0.467
Tangent contact point, virtual spur gear (mm)	[x,y]	1.627 /19.603	
Tangent contact point, virtual spur gear (mm)	[x,y]		1.696 /78.728
Diameter at critical cross section (mm)	[d_{sFn}]	39.342	157.493
Tangent at critical cross section (°)	[$α_{sFn}$]	30.000	30.000
Notch parameter	[q_s]	2.303	2.420
Helix angle factor	[Yβ]	0.967	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	21.00	20.00
Nominal stress at tooth root (N/mm²)	[σF0]	122.68	142.59
Tooth root stress (N/mm²)	[σF]	222.22	258.28
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.991	0.997
Surface factor	[YRrelT]	0.990	0.990
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	0.868	0.892
$Y_{δrelT}·Y_{RrelT}·Y_X·Y_{NT}$		0.852	0.880
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST}·σ_{Flim}$ (N/mm²)	[σFE]	840.00	726.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	550.39	491.26
Limit strength tooth root (N/mm²)	[σFG]	715.51	638.64

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.30	1.30
Safety for tooth root stress	[SF=σFG/σF]	3.22	2.47
Transmittable power (kW)	[kWRating]	30.96	23.78

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.379	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.798	

Helix angle factor	[Zβ]	1.015	
Effective facewidth (mm)	[beff]	20.00	
Nominal contact stress (N/mm²)	[σH0]	625.74	
Contact stress at operating pitch circle (N/mm²)	[σHw]	860.28	
Coefficient [fZCa] 1.20 (Helical gear sets without flank modifications)			
Single tooth contact factor	[ZB,ZD]	1.10	1.10
Contact stress (N/mm²)	[σHB, σHD]	942.39	942.39
Lubricant coefficient for N_L	[ZL]	0.988	0.988
Speed factor at N_L	[ZV]	0.981	0.981
Roughness factor for N_L	[ZR]	0.780	0.780
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.867	0.890
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.655	0.672
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	862.32	706.42
Pitting stress limit (N/mm²)	[σHG]	819.21	671.10

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	0.95	0.95
Safety factor for contact stress on operating pitch circle	[SHw]	0.95	0.78
Safety against pressure, σHG/σHBD Single contact	[SHBD]	0.87	0.71
Safety regarding transmittable torque	[SHBD²]	0.76	0.51
Transmittable power (kW)	[kWRating]	10.47	7.02

8 Micropitting

Calculation method according to	ISO/TS 6336-22:2018	
Lubricant load according to FVA Info sheet 54/7 10, Mobilgear 600 XP 150		
Reference data FZG-C Test:		
Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	128.037
Contact temperature (°C)	[θBRef,A]	253.806
Lubrication gap thickness (µm)	[hRef,A]	0.043
Specific film thickness in test	[λGFT]	0.085

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.120	
Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.106	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	1.831	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	78.410	
Tip relief factor	[XCa(A)]	1.817	
Loss factor	[HV]	0.105	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02051	
Dynamic viscosity (Ns/m²)	[ηtM]	23.768	
Roughness average value (µm)	[Ra]	3.000	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes		1		1
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)			1	
Diameter (mm)	[dy]	40.877		162.962
Relative radius of curvature (mm)	[ρred]		5.162	
Load sharing factor	[XY]		0.658	
X_Y interpolated between the values $X_Y(\varepsilon_b=0.8)$ and $X_Y(\varepsilon_b=1.2)$ according to ISO/TC60/WG6.				
Contact stress (N/mm²)	[pH]		689.325	
Contact stress (N/mm²)	[pdyn]		947.689	
Minimal specific film thickness	[λGFY]		0.061	(h_Y=0.183 µm)
Safety against micropitting	[Sλ(B)]		0.511	

9 Scuffing load capacity

Calculation method according to		ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------	
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780	
Relevant tip relief (µm)	[Ca]	5.00	5.00	
Optimal tip relief (µm)	[Ceff]		6.23	
C_a taken as optimal in the calculation. 0=no, 1=yes		1	1	
		---------- Pair ----------		
Helical load factor for scuffing	[KBγ]		1.240	
Lubrication coefficient for lubrication type	[XS]		1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),		12
Multiple meshing factor	[Xmp]		1.000	
Relative structural factor, scuffing	[XWrelT]		1.500	
Effective facewidth (mm)	[beff]		20.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]		183.228	
$K_{B\gamma}$ = 1.240, $w_{Bt}\cdot K_{B\gamma}$ = 227.133				
Angle factor	[Xαβ]		0.993	
ε_1 =0.917, ε_2 =0.654				

9.1 Flash temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]		0.848	
Tooth mass temperature (°C)	[θMi]		80.26	
$\theta_{Mi} = \theta_{oil} + X_S\cdot 0.47\cdot X_{mp}\cdot\theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]		11.20	
Scuffing temperature (°C)	[θS]		440.38	
Γ coordinates (point of highest temperature)	[Γ]		0.374	
[Γ.A]= -0.388, [Γ.E]= 0.544				
Maximum flash temperature (Δ°C)	[θflamax]		21.54	
Maximum contact temperature (°C)	[θBmax]		101.80	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]		50.058	
Approach factor	[XJ]		1.000	
Load sharing factor	[XΓ]		0.658	
Dynamic viscosity (mPa*s)	[ηM]		26.82	(75.0 °C)
Coefficient of friction	[μ_m]		0.085	
Required safety	[SBmin]		2.000	
Margin of safety for scuffing, flash temperature	[SB]		13.634	

9.2 Integral temperature-criteria

		---------- Pair ----------

Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	80.41	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	441.52	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.244	
Dynamic viscosity (mPa*s)	[ηOil]	26.82	(75.0 °C)
Mean coefficient of friction	[μ_m]	0.105	
Geometry factor	[XBE]	0.236	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.817	
Mean flash temperature (Δ°C)	[θflaint]	7.73	
Integral tooth flank temperature (°C)	[θint]	92.01	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	4.798	
Safety factor for transmitted moment (int.-T.)	[SSL]	21.544	

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 d24	DIN 3967 d25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.044 / -0.064	-0.080 / -0.130

10.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	13.000
Base tangent length (no backlash) (mm)	[Wk]	16.451	57.809
Base tangent length with allowance (mm)	[Wk.e/i]	16.410 /16.391	57.734 /57.687
(mm)	[ΔWk.e/i]	-0.041 / -0.060	-0.075 / -0.122
Diameter of measuring circle (mm)	[dMWk.m]	42.218	160.682

10.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	2.713	2.527
Effective diameter of ball/pin (mm)	[DMeff]	2.750	2.750
Radial single-ball measurement, no backlash (mm)	[MrK]	23.341	82.625
Radial single-ball measurement (mm)	[MrK.e/i]	23.294 /23.273	82.522 /82.458
Diameter of measuring circle (mm)	[dMMr.m]	42.672	161.251
Diametral measurement over two balls, no backlash (mm)	[MdK]	46.607	165.251
Diametral measurement over two balls (mm)	[MdK.e/i]	46.514 /46.471	165.045 /164.916
Diametral measurement over pins, no backlash (mm)	[MdR]	46.681	165.251
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	46.588 /46.545	165.045 /164.916
Measurement over 2 pins, free, AGMA 2002 (mm)	[dk2f.e/i]	46.508 /46.466	0.000 / 0.000
Measurement over 2 pins, transverse, AGMA 2002 (mm)	[dk2t.e/i]	46.660 /46.617	0.000 / 0.000
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	46.588 /46.545	165.045 /164.916

10.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	45.687	163.968
Reference chordal height from da.m (mm)	[hac]	2.015	1.605
Tooth thickness at height hac, chord (mm)	[sc]	2.710	2.437
(mm)	[sc.e/i]	2.667 / 2.647	2.357 / 2.307

Tooth thickness on reference circle, arc (mm)	[sn]	2.712	2.437
(mm)	[sn.e/i]	2.668 / 2.648	2.357 / 2.307

10.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.014 / -0.014
Radial backlash (mm)	[jrw.e/i]	0.276 / 0.147
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.215 / 0.115
Normal backlash (mm)	[jnw.e/i]	0.194 / 0.105
Center distance allowances (mm)	[Aa.e/i]	0.018 / -0.018
Backlash free center distance (mm)	[aControl.e/i]	101.680 / 101.587
Backlash free center distance, allowances (mm)	[jta]	-0.165 / -0.258

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	39.884	158.402
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.061	0.211
Tip clearance (mm)	[c0.i(aControl)]	0.227	0.177

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.1522 / 0.0814	(0°9'8" / 0°4'53")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.5864 / 0.3134	(0°35'11" / 0°18'48")

11 Gear tolerances

		Gear 1	Gear 2
According to DIN 3961:1978			
Accuracy grade	[Q-DIN3961]	6	6
Single pitch deviation (µm)	[fp]	±7.00	±8.00
Normal base pitch deviation (µm)	[fpe]	±7.00	±8.00
Adjacent pitch difference (µm)	[fu]	8.00	10.00
Total cumulative pitch deviation (µm)	[Fp]	19.00	29.00
Pitch span deviation (µm)	[Fpz/8]	±12.00	±18.00
Profile form deviation (µm)	[ff]	6.00	6.00
Profile slope deviation (µm)	[fHα]	±5.00	±5.00
Total profile deviation (µm)	[Ff]	8.00	8.00
Helix form deviation (µm)	[ffβ]	5.50	4.00
Helix slope deviation (µm)	[fHβ]	±9.00	±8.00
Total helix deviation (µm)	[Fβ]	10.00	9.00
Runout (µm)	[Fr]	14.00	19.00
Tooth Thickness Variation (µm)	[Rs]	8.00	11.00
Single flank composite, total (µm)	[Fi']	22.00	30.00
Single flank composite, tooth-to-tooth (µm)	[fi']	10.00	11.00
Radial composite, total (µm)	[Fi'']	17.00	24.00
Radial composite, tooth-to-tooth (µm)	[fi'']	6.00	10.00
According DIN 58405:1972 (Precision Mechanics)			
Axis alignment error (µm)	[fp]	±17.31	±17.31
Flank direction error (µm)	[fβ]	±5.00	±5.00
Runout (µm)	[Trk, Fr]	21.00	28.00
Composite error (µm)	[Fi'']	20.00	25.00
Tooth-to-tooth composite error (µm)	[fi'']	7.00	9.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99		
Total helix deviation for calculation (µm)	[Fβ]	10.00
Distance between bearings for calculation (mm)	[LG]	42.00

Maximum value for axis deviation error (µm)	[fΣβ]	± 10.50
Maximum value for axis inclination error (µm)	[fΣδ]	± 21.00

12 Modifications and determination of the tooth form

12.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, linear

Caa = 5.000 µm LCa = 0.865 ·mn dCa = 44.403 mm

Gear 2

Symmetric (both flanks)

- Tip relief, linear

Caa = 5.000 µm LCa = 0.865 ·mn dCa = 162.962 mm

12.1.1 Tip relief verification

		------- Gear 1	---------- Gear 2 -------
Diameter (mm)	[dcheck]	45.652	163.933
Tip relief left/right (µm)	[Ca L/R]	4.9 / 4.9	4.8 / 4.8
Diameter (mm)	[dNa.i]	45.682	163.963
Tip relief left/right (µm)	[Ca L/R]	5.0 / 5.0	5.0 / 5.0

12.2 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	102.821

13.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	0.231	0.537
Total mass (kg)	[Σm]	0.768	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	5.183e-05	0.00317
System (da+df)/2...di (kg*m²)	[J]	0.0002655	

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	0.126
Torsion when subjected to nominal torque (°)	[δcr]	0.018
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	61.793

13.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.101
Wear sliding coefficient by Niemann	[ζw]	0.697
Loss factor	[HV]	0.105
Gear power loss (kW)	[PVZ]	0.133
Meshing efficiency (%)	[ηz]	98.937

13.4 Indications for the manufacturing by wire cutting

		------- Gear 1	---------- Gear 2 -------
Deviation from theoretical tooth trace (µm)	[WireErr]	148.8	38.7
Permissible deviation (µm)	[Fb/2]	5.0	4.5

14 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.3000
Required safety for tooth flank	[S_{Hmin}]	0.9500
Required service life	[H]	20000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	16.8997

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	190.7	16.9

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 20000.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	0.0000	9999.9999	9999.9999

Damage relative to the system service life (H_{att}, 16.9 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	0.0000	8.8601	100.0000

15 Reliability calculation

15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

15.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	177000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	177000	1.300e+00	3.043e+07	1.450e+08	0.0000
2	Tooth root	45952	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	45952	1.300e+00	7.000e+05	3.335e+06	0.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

15.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	20000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	0.0000
Reliability R, gears subsystem (%)	[R_{subG}]	0.0000
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	16.8997
Service life H, gears subsystem (h)	[H_{subG}]	16.8997

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

16.2 Calculations and factors

- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with Z_β = 1/(COS(β)^0.5)
- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

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